

Nineteenth-century British physicist James Clerk Maxwell's theory of electromagnetism proved that electricity, magnetism, and light are all different aspects of the same underlying phenomenon: an idea that remains of fundamental importance.

MILESTONES

FIRST SCIENCE PAPER

"Oval Curves," published in 1846, arises from his childhood love of drawing using pins and thread.

PIVOTAL THEORY

In 1864, he publishes "A Dynamical Theory of the Electromagnetic Field," with detailed equations.

FINAL POST

Returns to Cambridge in 1871 and becomes the first Cavendish Professor of Experimental Physics.

PUBLISHES KEY WORK

In 1873, his two-volume *Treatise on Electricity and Magnetism* explains his electromagnetic work.

James Clerk Maxwell was an only child, the son of a wealthy Edinburgh lawyer. His mother, who tutored him at home in his early years, died when he was 8 years old. The young Maxwell was extremely clever and inquisitive and intrigued by how things work. At the age of 14, he published his first scientific paper, on the subject of geometry. He also memorized the Bible and remained a devout Protestant all his life.

Maxwell went to Edinburgh University at the age of 16 to study physics (then called "natural philosophy") and mathematics. Finding the taught courses unchallenging, he began to conduct his own experiments in his spare time and published two more papers. Aged 19, he went to study mathematics at the University of Cambridge, and 5 years later became a Fellow of Trinity College. His intellectual interests were many and varied: in 1855, he published "Experiments on color as perceived by the eye." He later created the first color photograph.

The rings of Saturn

At 25, Maxwell took up a professorship at the University of Aberdeen. He also immersed himself in a challenge that had been set for the Adams Prize, a prestigious mathematics prize at Cambridge: to explain Saturn's rings. Scientists could not understand how the rings could

The first-ever color photograph, of this tartan ribbon, was produced by Maxwell in 1861. He made it using red, green, and blue color filters.



"Every atom of creation is unfathomable in its **perfection."**

James Clerk Maxwell, 1873